



Data Cleaning



Take home message: “there is probably a function for that”

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Aims of the session – Get a glimpse of real data

So far, you've learned how to manipulate data and visualise it. Today we focus on what happens when data aren't tidy



<https://becominghuman.ai/day-7-data-cleaning-all-that-you-need-to-know-about-it-23b05738abe7>

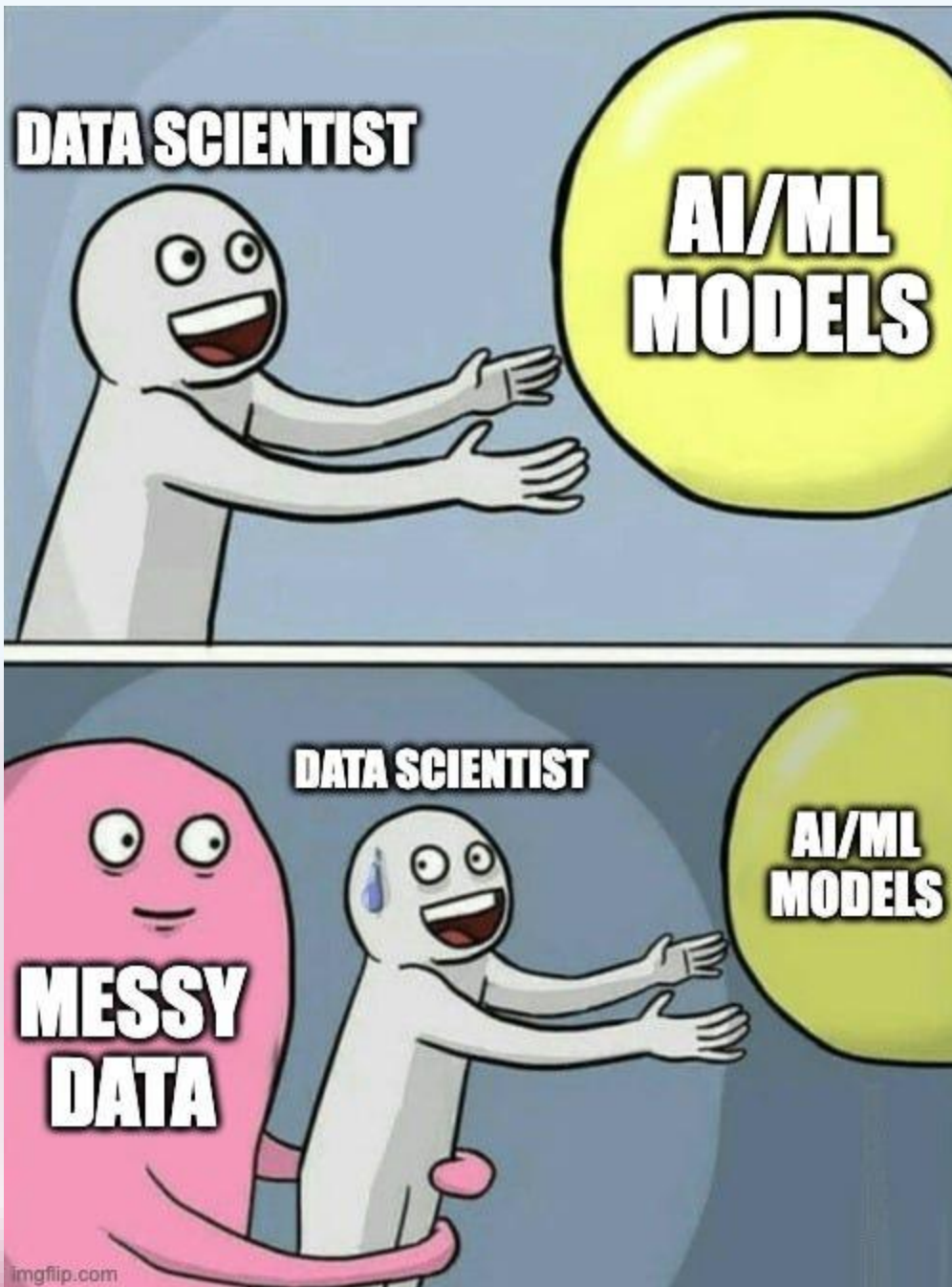
Take home message: “there is probably a function for that”

What is wrong with these datasets?

```
# A tibble: 10 × 3
```

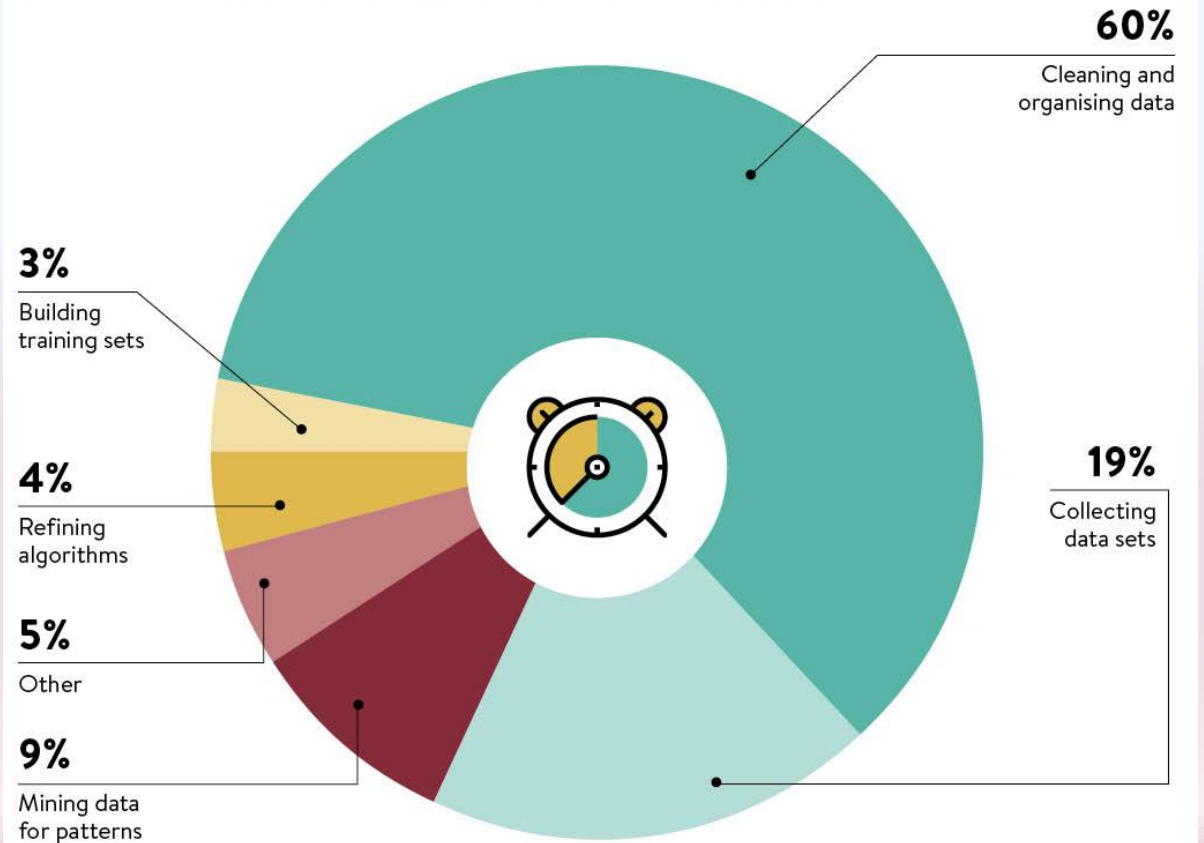
	month	x	y	date
	<chr>	<chr>	<chr>	<chr>
1	"January"	"2.21"	"1"	01/01/2025
2	"Jan@ua^ry"	"2.29"	"1"	01/01/2025
3	"January "	"1.46"	"0"	01/01/2025
4	"FEBRUARY"	"0.55"	"1"	01/02/2025
5	".February"	"0.06"	"0"	2025/02/01
6	"Februa(ry"	<NA>	"1"	01/02/2025
7	"March"	"0.11"	<NA>	2025/03/01
8	"Ma!rch"	"0.13"	"1"	01/03/2025
9	"April"	"0.63"	"1"	01/04/2025
10	"April"	<NA>	"0"	2025/04/01

color	director_name	duration	gross	movie_title	language	country	budget	title_year	imdb_score
Color	Martin Scorsese	240	116866727	The Wolf of Wall StreetÃ	English	USA	100000000	2013	8.2
Color	Shane Black	195	408992272	Iron Man 3Ã	English	USA	200000000	2013	7.2
color	Quentin Tarantino	187	54116191	The Hateful EightÃ	English	USA	44000000	2015	7.9
Color	Kenneth Lonergan	186	46495	MargaretÃ	English	usa	14000000	2011	6.5
Color	Peter Jackson	186	258355354	The Hobbit: The Desolation of SmaugÃ	English	USA	225000000	2013	7.9
	N/A	183	330249062	Batman v Superman: Dawn of JusticeÃ	English	USA	250000000	202	6.9
Color	Peter Jackson	-50	303001229	The Hobbit: An Unexpected JourneyÃ	English	USA	180000000	2012	7.9
Color	Edward Hall	180		RestlessÃ	English	UK		2012	7.2
Color	Joss Whedon	173	623279547	The AvengersÃ	English	USA	220000000	2012	8.1
Color	Joss Whedon	173	623279547	The AvengersÃ	English	USA	220000000	2012	8.1
	Tom Tykwer	172	27098580	Cloud AtlasÃ	English	Germany	102000000	2012	-7.5
Color	Null	158	102515793	The Girl with the Dragon TattooÃ	English	USA	90000000	2011	7.8
Color	Christopher Spencer	170	59696176	Son of GodÃ	English	USA	22000000	2014	5.6
Color	Peter Jackson	164	255108370	The Hobbit: The Battle of the Five ArmiesÃ	English	New Zealand	250000000	2014	7.5
Color	Tom Hooper	158	148775460	Les MisÃrablesÃ	English	USA	61000000	2012	7.6
Color	Tom Hooper	158	148775460	Les MisÃrablesÃ	English	USA	61000000	2012	7.6



Data cleaning takes muuuch more time than expected

WHAT DATA SCIENTISTS SPEND THE MOST TIME DOING



Source: CrowdFlower 2016

Data Cleaning Techniques

01 Remove
duplicates

02 Detect &
remove Outliers

03 Remove
irrelevant data

04 Standardize
capitalization

05 Convert
data type

06 Clear
formatting

07 Fix
errors

08 Language
translation

09 Handle
missing values

What is a factor?

Let's play: Is it a factor?

Treatment	Chlorophylls
WellWatered	23
WellWatered	21
WellWatered	25
WellWatered	24
Drought	10
Drought	8
Drought	9
Drought	7

Size	Body mass
Small	10
Small	15
Medium	25
Medium	30
Large	50
Large	60

Sex	Size
F	12
M	18
F	25
M	35
F	45
M	55

Zone	Density
1	5
1	8
1	7
2	3
2	5
2	4
3	10
3	15
3	17

ID
1
2
3
4
5
6
7
8

Day of the week, postcode,
life stage,
presence/absence,
temperature, age, year

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3	15
3	17



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Day of the week, postcode, life stage, presence/absence, temperature, age, year



Depends on the context

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2	5
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Day of the week, postcode,
life stage,
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In R programming, a **factor** is a data object used to store categorical variables, representing a fixed set of unique, known values (levels)

In plain language, a **grouping variable**.

A variable should be a factor when:

- It represents **groups or categories**
- Arithmetic operations do **not make sense**
- The numeric appearance (if any) is just a label

Why is it important to declare a factor a factor in R?

```
df$treatment <- factor(df$treatment)
```

If a variable is **numeric**, R assumes:

- There is a meaningful numeric scale
- The relationship is linear
- The distance between values matters

If a variable is a **factor**, R:

- Treats each level as a separate group
- Compares group means
- Creates dummy variables internally



Models behave differently
Plots change
Summaries behave differently
Ordering matters



Forcats to handle factors

forcats helps you manage categorical variables in a **clear, explicit, and reproducible way**. Three ways to declare factors:

```
months <- as.factor(x1)
months <- factor(x1, levels = month_levels)
months <- fct(x1, levels = month_levels)
```

Reorder levels, collapse levels, declare transformations of labels in code

`fct_reorder()` → order by a numeric variable

`fct_relevel()` → manually set order

`fct_lump()` → combine rare categories

`fct_infreq()` → order by frequency

`fct_rev()` → *Reverses the order of factor levels.*

`fct_recode()` → *Renames factor levels.*

...

You don't need to remember all *forcats* functions, but you now know where to look

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U2	SL1	SL2	SU1	SU2
----	-----	-----	-----	-----

Site	
ML1	
ML1	
ML1	
ML1	
ML1	
ML1	

ML1	ML2	MU1	MU2	PL2	PU2	SL1	SL2	SU1	SU2
-----	-----	-----	-----	-----	-----	-----	-----	-----	-----

Reshaping your data



table4a

country	1999	2000
A	0.7K	2K
B	37K	80K
C	212K	213K



country	year	cases
A	1999	0.7K
B	1999	37K
C	1999	212K
A	2000	2K
B	2000	80K
C	2000	213K

pivot_longer(data, cols, names_to = "name",
values_to = "value", values_drop_na = FALSE)

"Lengthen" data by collapsing several columns
into two. Column names move to a new
names_to column and values to a new values_to
column.

```
pivot_longer(table4a, cols = 2:3, names_to = "year",  
values_to = "cases")
```

table2

country	year	type	count
A	1999	cases	0.7K
A	1999	pop	19M
A	2000	cases	2K
A	2000	pop	20M
B	1999	cases	37K
B	1999	pop	172M
B	2000	cases	80K
B	2000	pop	174M
C	1999	cases	212K
C	1999	pop	1T
C	2000	cases	213K
C	2000	pop	1T



country	year	cases	pop
A	1999	0.7K	19M
A	2000	2K	20M
B	1999	37K	172M
B	2000	80K	174M
C	1999	212K	1T
C	2000	213K	1T

pivot_wider(data, names_from = "name",
values_from = "value")

The inverse of pivot_longer(). "Widen" data by
expanding two columns into several. One column
provides the new column names, the other the
values.

```
pivot_wider(table2, names_from = type,  
values_from = count)
```

unnest_longer(data, col, values_to = NULL, indices_to = NULL)
Turn each element of a list-column into a row.

```
starwars |>  
select(name, films) |>  
unnest_longer(films)
```

name	films
Luke	<chr [5]>
C-3PO	<chr [6]>
R2-D2	<chr [7]>



name	films
Luke	The Empire Strik...
Luke	Revenge of the S...
Luke	Return of the Jed...
C-3PO	The Empire Strik...
C-3PO	Attack of the CL...
C-3PO	The Phantom M...
R2-D2	The Empire Strik...
R2-D2	Attack of the CL...
R2-D2	The Phantom M...

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Handling (not eliminating) missing values

What is a missing value?

Is not a zero!

NA = Not Available

Some people might code NAs as NaN, NULL or 0

Some functions will not work with NA

```
mean(x) # returns NA  
mean(x, na.rm=TRUE) # returns 2
```

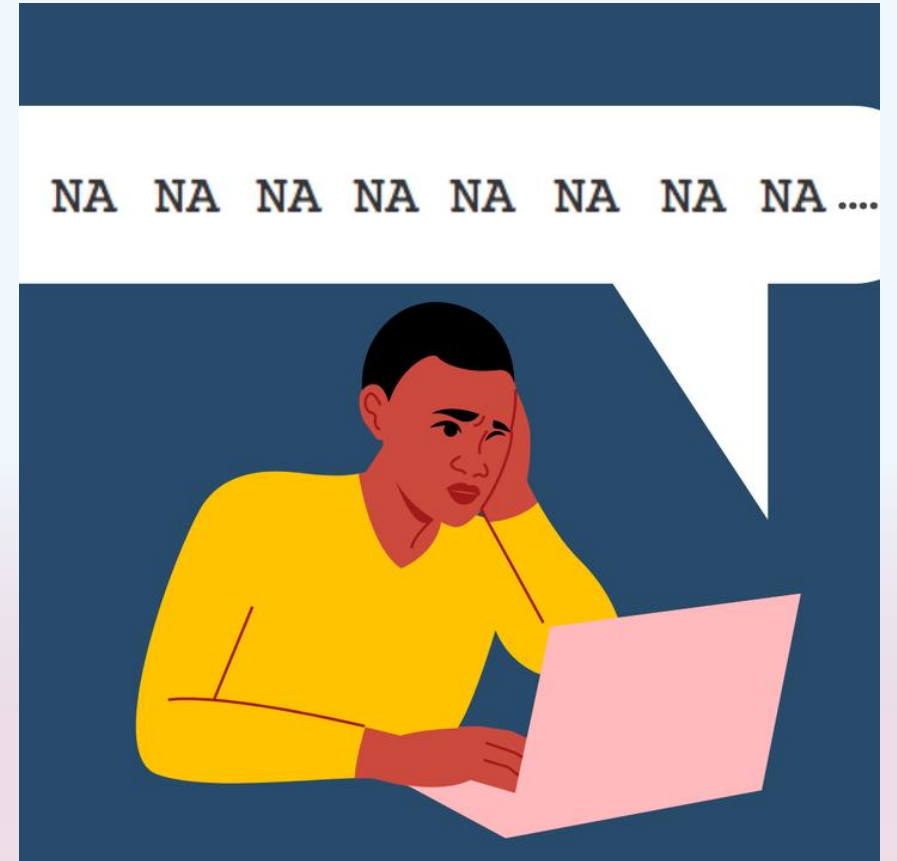
Some analysis ignore NA by default

```
lm(body_mass_g ~ species, data = penguins)
```

`is.na()` will work on individual values, vectors, lists, and data frames. It will return **TRUE** or **FALSE** where you have an **NA** or where you don't.

You can also combine `is.na()` with `sum()` and `which()` to figure out how many **NA**s you have and where they're located.

```
sum(is.na(variable))
```



<https://www.rforecology.com/post/introduction-to-missing-data-in-r/>

Handling missing values

- Omit them:

- `na.omit(variable)`
- `dataset[complete.cases(dataset), ]`
- `drop_na(dataset, variables)`

- Replace them:

With the mean, median, mode

K-nearest neighbors (knn) or hot-deck

Multivariate interpolation

Random number

We might lose a lot of data! A missing value in a row might delete the whole row.

X1	X2	X3
NA	4	1
7	1	4
8	NA	2
9	NA	9
3	4	6



Your turn!

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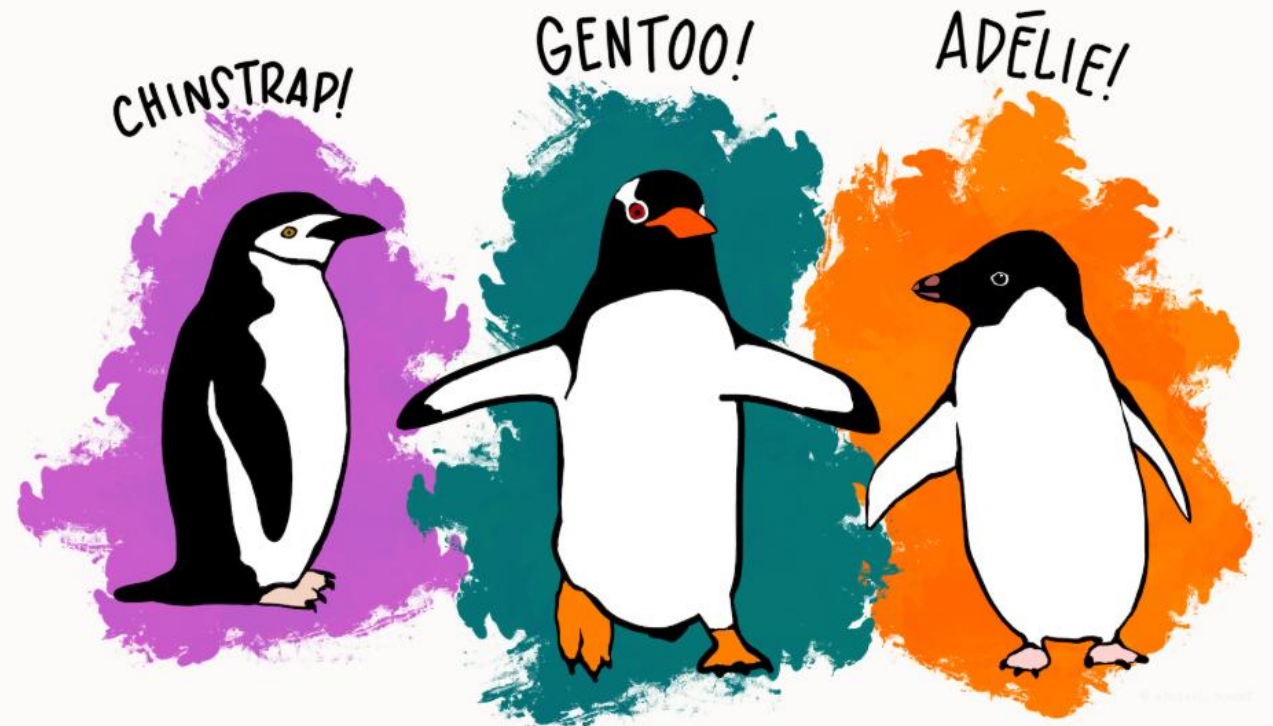
09 Handle missing values

Your turn!

The palmerpenguins data contains size measurements for three penguin species observed on three islands in the Palmer Archipelago, Antarctica. These data were collected from 2007 - 2009 by Dr. Kristen Gorman with the [Palmer Station Long Term Ecological Research Program](#), part of the [US Long Term Ecological Research Network](#). The data were imported directly from the [Environmental Data Initiative](#) (EDI) Data Portal, and are available for use by CC0 license (“No Rights Reserved”) in accordance with the [Palmer Station Data Policy](#).

Play the “Data cleaning challenge”

Download the Qmd from the One Drive folder



The Palmer Archipelago penguins. Artwork by @allison_horst.